CLASSIFICATION C1 - INTERNAL

AAKASH KARUNAKARA SHETTY

Full Stack Developer | Generative AI & Agentic Systems | Python / FastAPI Backend

Mumbai, India | +91 80975 23767 | aakashshetty227@gmail.com

linkedin.com/in/aakash-shettys | github.com/aakash-shetty | aakash-shetty.vercel.app

EDUCATION

M.Sc. Business Analytics (Data Analytics) — Manipal Academy of Higher Education • GPA 8.05/10 **2023 – 2025**

B.Com — KJ Somaiya College of Arts & Commerce, Mumbai University • GPA 8.82/10 **2020 – 2023**

Coursework: Machine Learning • NLP • Statistical Methods • Data Mining & Warehousing • Deep Learning • Time Series • Big Data • DBMS

TECHNICAL SKILLS

GenAI & Agents: AWS Bedrock (Claude), OpenAI GPT-4o/4.1, LangChain, MCP, agents, RAG/GraphRAG, hybrid search, text-to-SQL, LLM evals
Backend: Python, FastAPI, async Python, REST APIs, SQLAlchemy, Pydantic, Node.js, Express.js, OAuth 2.0, JWT, Azure AD SSO
ML & Data: scikit-learn, gradient boosting, time-series forecasting, feature engineering, pandas, NumPy, ETL pipelines, OCR, OpenCV
Data & Cloud: PostgreSQL (AWS RDS), pgvector, Redshift, MySQL; AWS (EC2, S3, RDS, CloudFront, Bedrock), Docker, Nginx, Linux, Git, CI/CD
Frontend: React 18, TypeScript, JavaScript, Redux, Next.js, SSE streaming, HTML/CSS

PROFESSIONAL EXPERIENCE

Allcargo Global Ltd. / ECU Worldwide — Data Science **Mumbai, India**

Data Scientist **Jul 2025 – Present**

ELMA 2.0 — Enterprise AI Assistant (Agentic RAG) | *Python, FastAPI, LangChain, MCP, pgvector, AWS Bedrock*

- Cut response time **~90%** (30–60s → 2–5s) for **3,000+** employees in India, USA and Canada; 1 of **3** engineers; 26-tool MCP server.
- Built the OCR pipeline indexing **1,500+** internal documents, searchable in plain English.

Pendulum AI — Freight Analytics & Natural-Language Q&A | *React 18, FastAPI, PostgreSQL/pgvector, AWS Bedrock*

- GraphRAG + hybrid retrieval over **1.6M+** shipment records via **43** FastAPI endpoints (**3,300**-node graph).
- Sole frontend owner (React 18, SSE chat, Azure AD SSO); adopted by **25** branch managers.

KAIROS & FRAME — Finance Q&A Agents & Email Triage | *Python, FastAPI, AWS Bedrock, pgvector, Microsoft Graph*

- Built **7** text-to-SQL agents over **3** production databases: P&L/EBIT answers in seconds vs. **2-3 days**.
- FRAME email triage (Microsoft Graph → Claude → Outlook) routes **~200** emails/day at **91%** accuracy.

ACE — Invoice Translation Platform | *FastAPI, GPT-4o-mini, PostgreSQL, AWS*

- Sole backend developer, live in ECU Cambodia: **100+** invoices/month, **25 → 3 min** each, never alters amounts or IDs.

RIVA — Dangerous-Goods Compliance Automation | *Python, GPT-4.1, OpenCV, Tesseract*

- Technical lead: validates **100+** live shipping documents across **14** IMDG categories (OCR, OpenCV, GPT-4.1); mentored **1** junior.

Data Science Intern **Jan 2025 – Jun 2025**

GLIMPSE — AI Sales Research Agent | *Python, FastAPI, LangChain, AWS Bedrock, PostgreSQL*

- Sole backend developer; used by **~900** sales reps: 13-section briefs in **~0.5 min** vs. **3+ hrs** manually; converted to full-time after internship.

PERSONAL PROJECTS

Verity — AI Support Resolution Agent | *Python, FastAPI, LangGraph, MCP, PostgreSQL, React/TypeScript*

- AI support agent where the model proposes and code authorizes every action (**4** risk tiers, audit log): **100%** correct outcomes and **0** unsafe actions on **100** held-out cases vs. **44%** for policy search alone.
- Shipped as a product: React/TypeScript app, MCP server (**7** tools), **600+** backend tests, CI, Docker.

Vanguard — Multi-Agent Tender Response System | *Python, FastAPI, LangGraph, pgvector, MCP, React/TypeScript*

- Multi-agent tender (RFP) responder with human approval: on a **50**-question dev tender (single run) caught **100%** of planted traps, **0** unsafe answers vs. **2** for a single agent; **97.5%** recall@5.
- FastAPI + pgvector backend with live SSE progress and audit log; React/TypeScript app; **300+** tests.

CERTIFICATIONS

- Business Analyst, **8-month** course on Advanced Excel, Power BI, Tableau, SQL, Python and R from ExcelR Solutions (USA), 2023.
- Smart Professional MIS, **10-month** course on MS Access, MS Excel, MS Word, PowerPoint, Automated Financial Accounting (Tally) and Automation in Excel with VBA, from Aptech Learning, 2022.

EXTRA-CURRICULAR ACTIVITIES

- College Placement Cell Representative, 2022-23: organized **2** placement drives securing **11** internships & **20+** job placements.
- Department of Lifelong Learning & Extension (DLLE) student representative at KJSCAC, 2022-23.